Introduction to Probability Theory and Statistics
Exam, June 2019

Probability

Problem 1
Two balanced dice are rolled.

(a) What is the probability that the sum of the two numbers that appear will be odd?

(b) What is the expected value of the sum of the two numbers that appear?

(c) What is the variance of the sum of the two numbers that appear?

(d) What is the probability that the product of the two numbers that appear will be odd?

(e) What is the expected value of the product of the two numbers that appear?

Problem 2
Three different robots R_1, R_2 and R_3 produce similar items. Robot R_1 makes 20% of all manufactured items; R_2 makes 30% and R_3 makes 50%. Suppose that 1% of the items produced by the first robot R_1 are defective; 2% of the items produced by the second robot R_2 are defective and 3% of the items produced by the third robot R_3 are defective.
Suppose that one item is selected at random from a batch of all produced items and it is found to be defective. Determine that probability that this item has been produced by R_2.

Problem 3
The joint pdf of two continuous random variables X and Y is as follows:

$$f(x,y) = \begin{cases} cxe^{-y} & 0 \le x \le 1, 0 < y < \infty \\ 0 & otherwise \end{cases}$$

(a) Find constant c.

(b) Find the marginal pdf for r.v. X, $f_X(x)$.

(c) Are X and Y independent?

(d) Find probability of the event that X is smaller than $1/2$ and Y is smaller than 1, that is, find $P(X < 1/2, Y < 1)$.

(e) Find $P(X = 1/2, Y = 1)$ and $P(X = 1/2, Y < 1)$.

(f) Find cumulative distribution function for r.v. X, $F_X(x)$.

(g) Draw cdf $F_X(x)$. What is the value of $F_X(-1)$ and value of $F_X(2)$?